

Fig. 9: Actual-size, single-side PCB for the relay driver section (on the right side of dotted line in Fig. 4)

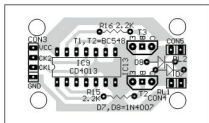


Fig. 10: Component layout for the PCB in Fig. 9

- Up to 9.6kbps data rate
- +9dBm output power (a range of about 200 metres)
- SAW-based architecture
- For the antenna, a 45cm wire is adequate

The output power and current drain for V_{CC} of 5V and 12V, respectively, are shown in Table I.

Pin configurations of the transmitter and receiver modules are shown in Figs 1 and 2, respectively. Technical specifications of the receiver module are shown in Table II.

Transmitter. Fig. 3 shows the transmitter circuit for PC-based wireless appliance control. The receiver address to be transmitted can be set with the help of an 8-way DIP switch (DIP-SW1). If any switch is open the pin connected to that switch is at logic 1, and if it is closed the respective pin is at logic 0. The data pins are pulled high via resistors R2 through R5.

When pin 2 of the parallel port

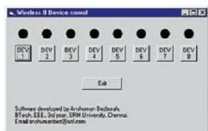


Fig. 11: Main screen on PC

goes high, the internal LED of the optocoupler (MCT2E) glows to drive the internal phototransistor into saturation and its pin 5 goes low.

Pin 10 (A8) of HT12E goes low through pin 5 of MCT2E and a '0' is sent at that data position, while other data pins represent the logic-1 state. The logic circuitry at the receiver-decoder end decodes the data appropriately for controlling the switch of an appliance.

Receiver-cum-decoder. Fig. 4 shows the receiver circuit for PC-based wireless appliance control. Assuming that an identical address is selected on the encoder and decoder, when any one of data pins D0 through D3 of the PC's parallel port on the transmitter is high, the corresponding data pin of the demodulator goes low at the decoder (HT12D). The data outputs (D8 through D11) of HT12D are connected to inverters N1 through N4. The output of inverters N1, N2 and N3 are connected to address inputs A, B and C of decoder 74LS138, respectively. The outputs of inverters N4 and N5 enable the decoder to drive the flip-flop (CD4013).

IC CD4013 is configured as a toggle flip-flop. The output of the flip-flop (IC9(A)) drives transistor T2 into saturation and relay RL1 energises. D7 through D14 are used as free-wheeling diodes. The relay contacts are used for

appliances.

When any data is received, the valid transmission pin (V_T) goes high to drive the transistor into saturation and LED1 lights up.

Construction

An actual-size, single-side PCB for the transmitter circuit (Fig. 3) is shown in Fig. 5 and its component layout in Fig. 6. Actual-size, single-side PCBs for the receiver-cum-logic control section (on the left side of dotted line in Fig. 4) and relay driver section (on the right side of dotted line in Fig. 4) are shown in Figs 7 and 9 and their component layouts in Figs 8 and 10, respectively. Connectors have been provided on the PCB for all the outputs (CK1 through CK8) of IC8, V_{CC} and GND. Thus it can be extended for the relay driver section at any desired location. The PCB of the relay driver is made for two relays to drive two appliances. You can extend the same to control more than two appliances by adding another PCB as shown in Fig. 9. Wire the circuit on the PCB as shown in Fig. 9 and extend it to any other appliance in the house.

Use bases for all the ICs. Check continuity between respective connections using a multimeter. Connect the electrical appliances through relay contacts.

Software

The source program for PC-based wireless appliance control is written in Visual Basic. To use the software on Win XP platform, the input32.dll file should reside in the same folder as the .exe file of the program. The main screen for PC-based wireless appliance control is shown in Fig. 11.

Note. The source code and other relevant files of this article are included in this month's EFY-CD.

FORM1.FRM

```
VERSION 5.00
Begin VB.Form Form1
Caption = "Wireless 8 Device control"
ClientHeight = 3480
ClientLeft = 60
ClientTop = 450
ClientWidth = 6090
LinkTopic = "Form1"
ScaleHeight = 3480
```

```
ScaleWidth = 6090
StartUpPosition = 3 'Windows Default
Begin VB.CommandButton Command10
Caption = "Exit"
Height = 375
Left = 2280
TabIndex = 9
Top = 1800
Width = 1215
```

```
End
Begin VB.CommandButton Command8
Caption = "DEV8"
Height = 495
Left = 5280
TabIndex = 7
Top = 840
Width = 495
End
Begin VB.CommandButton Command7
```